

Well-quasi-ordering certain classes of infinite graphs

Bachelor Thesis Defense

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Definition (Quasi-order)

A binary relation $\preceq$ on a class Q is a **quasi-order** if it is reflexive and transitive.

- $x \equiv y$ iff $x \preceq y$ and $y \preceq x$ x and y are **equivalent**
- $x \prec y$ iff $x \preceq y$ and $x \not\equiv y$.
- x is **minimal** if there is no $y \prec x$.

Definition (Well-quasi-order)

Q is **wqo** if in every infinite sequence $x_1, x_2, x_3, \dots$ of elements of Q , there are $i < j$ such that $x_i \preceq x_j$.

Well-quasi-orderings

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Proposition

The following conditions are equivalent to Q being wqo:

- 1 Every infinite sequence $x_1, x_2, x_3, \dots$ of elements of Q contains an **infinite non-decreasing subsequence** $x_{i_1} \preceq x_{i_2} \preceq x_{i_3} \preceq \dots$ for some indices $i_1 < i_2 < i_3 < \dots$.
- 2 Q admits no **infinite decreasing sequences** $x_1 \succ x_2 \succ x_3 \succ \dots$ and no **infinite antichains**.
- 3 Every nonempty subset of Q contains **at least one** but only **finitely many** non-equivalent **minimal** elements.

Definition

A property φ of Q is **monotone** if whenever $\varphi(x)$ holds and $y \preceq x$, then $\varphi(y)$ holds as well. It is **finitely testable** if there exists a finite **obstruction set** $\mathcal{F}$ such that for all $x \in Q$ we have

$$\varphi(x) \text{ holds} \iff (\nexists z \in \mathcal{F}) : z \preceq x.$$

Example (Wagner's Theorem)

Planarity is a finitely testable property of **finite graphs** with respect to the **minor** relation. Its minimal obstructions are K_5 and $K_{3,3}$.

Proposition

If Q is wqo, then every monotone property of Q is finitely testable.

Proof sketch: Define $\mathcal{F}$ as the (nonempty finite) set of non-equivalent minimal elements of the class

$$\{x \in Q \mid \varphi(x) \text{ does not hold}\}.$$

The Graph Minor Theorem

Proposition

If Q is wqo, then every monotone property of Q is finitely testable.

Theorem (Robertson, Seymour, 1983–2004)

*The class of all **finite graphs** is **wqo** by the **minor** relation.*

Corollary

For every surface, there exists a finite obstruction set for embeddability of finite graphs on that surface.

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For every surface, there exists a finite obstruction set for embeddability of finite graphs on that surface.

Theorem (Thomas, 1988)

*The class of all **finite or infinite graphs** is **not wqo** by the **minor** relation.*

Conjecture: The class of all **countable** graphs is wqo by the minor relation.

Theorem (Komjáth, 1995)

For every $\kappa > \omega$, there is an antichain of 2^κ many graphs of size κ (with respect to the minor relation).

Better-quasi-orderings

Theorem (Kruskal's Tree Theorem, 1960)

*The class of all **finite trees** is **wqo** by the **topological minor** relation.*

In order to extend this result to infinite trees, Nash-Williams introduced the stronger notion of **better-quasi-orderings** or **bqo**. The wqos encountered in practice are, in fact, bqos.

- Well-ordered $\implies$ bqo $\implies$ wqo.

Theorem (Nash-Williams, 1965)

*The class of all **finite or infinite trees** is **bqo** by the **topological minor** relation.*

Better-quasi-orderings

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Theorem (Nash-Williams, 1965)

*The class of all **finite or infinite trees** is **bqo** by the **topological minor** relation.*

Theorem (Thomas, 1989)

*Any class of **finite or infinite graphs** with **bounded tree-width** is **bqo** by the **minor** relation.*

Corollary

*Any class of **finite or infinite graphs** with **bounded tree-depth** is **bqo** by the **minor** relation.*

Proof: Bounded tree-depth implies bounded tree-width.

Tree-depth & Ding's Theorem

Theorem (Ding, 1992)

Any class of *finite graphs* with *bounded tree-depth* is *wqo* by the *induced subgraph* relation.

- *Tree-depth* measures how close a given graph is to a *star*.

Theorem (J.S.)

Any class of *finite or infinite graphs* with *bounded tree-depth* is *bqo* by the *induced subgraph* relation.

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Tree-depth of finite graphs was studied by Nešetřil and Ossona de Mendez. I extended the notion to infinite graphs and showed that its basic properties hold in this more general setting. For example:

Lemma

- If H is a minor of G , then $\text{td}(H) \leq \text{td}(G)$ true even when $\text{td}(H) \geq \omega$
- If G has finite tree-depth, then $\text{tw}(G) < \text{td}(G)$.

I also established a *compactness* theorem:

Theorem (J.S.)

Let G be an arbitrary graph. Then $\text{td}(G) < k$ if and only if $\text{td}(H) < k$ for every *finite* subgraph H of G .

Bounded tree-depth classes

Definition

The **path-depth** of a graph G is $\max\{n \mid P_n \subseteq G\}$, or ∞ if the maximum does not exist.

Example

- path-depth of a tree is $1 +$ its diameter.
- path-depth of K_n is n .

Proposition

A class of graphs $\mathcal{G}$ has **bounded tree-depth** if and only if it has **bounded path-depth**.

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A class of graphs $\mathcal{G}$ has **bounded tree-depth** if and only if it has **bounded path-depth**.

Theorem (J.S.)

Any class of **finite or infinite graphs** with **bounded path-depth** is **bqo** by the **induced subgraph** relation.

Corollary

Any class of **finite or infinite trees** with **bounded diameter** is **bqo** by the (induced) **subgraph** relation.

Definition (Rooted tree of height at most ω)

An (order-theoretic) **tree** is a partially ordered set $(T, <_T)$ such that for every $x \in T$, the **downset**

$$(\leftarrow, x) := \{y \in T \mid y <_T x\}$$

is a **finite chain**, and there is a unique minimal element called the **root** of T .

Terminology:

- $|x|_T := |(\leftarrow, x)|$ is the **height** or **level** of x in T .
- $x \wedge y$ is the **meet** of x and y (their closest common ancestor).
- A **branch** of T is an inclusion-maximal chain $B \subseteq T$.

Definition (Embeddings of trees)

An **embedding** of S into T is a mapping $\varphi: S \rightarrow T$ such that $x <_S y \implies \varphi(x) <_T \varphi(y)$. It is a

- 1 **level-embedding** if $|x|_S = |\varphi(x)|_T$ holds for all $x \in S$.
- 2 **homeomorphic embedding** if it is injective and $\varphi(x \wedge y) = \varphi(x) \wedge \varphi(y)$ for all $x, y \in S$.

Containment relations of trees

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Intuition:

- Level-embeddings are **homomorphisms** of rooted trees (edges go to edges).
- Injective level-embeddings are **subgraphs** for rooted trees where **root** goes to **root**.
- Homeomorphic embeddings are **topological minors** for rooted trees.

Proposition

Let S and T be trees. Then the two following conditions are equivalent:

- ① There exists an **embedding** of S into T .
- ② There exists a **level-embedding** of S into T .

Tree-homomorphisms

Theorem (Nash-Williams, 1965)

*The class of all order-theoretic trees is wqo by the **homeomorphic embedding** relation.*

Corollary

*The class of all order-theoretic trees is wqo by the **level-embedding** relation.*

The proof of Nash-Williams' theorem relies on the heavy machinery of bqo theory and is highly intricate. I found a new, direct proof of this corollary that completely avoids bqo theory.

Proof idea:

- Observe that it is trivial for trees of finite height and for trees that contain infinite branches.
- Hence it is enough to focus on trees of infinite height but without an infinite branch.
- Define a hierarchy of such trees.
- Prove by transfinite induction along this hierarchy that for every ordinal α , the class of all trees that appear before level α is wqo by level-embeddings.
- Show that every tree without an infinite branch appears in some level of the hierarchy.

Generalizations of WQOs

Definition (Shelah, 1982)

Q is κ -wqo (where κ is a cardinal) if in every sequence $\langle q_\alpha \mid \alpha < \kappa \rangle$ there are $\alpha < \beta$ such that $q_\alpha \preceq q_\beta$. The **well-ordering number** of Q is the least cardinal κ such that Q is κ -wqo (if it exists).

Example

- **Finite graphs** are ω_1 -wqo by [your favorite graph containment relation].
- The class $\mathcal{G}^{\leq \kappa}$ of **graphs of size $\leq \kappa$** is $(2^\kappa)^+$ -wqo by [your favorite graph containment relation].

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Theorem (Vopěnka, Pultr, Hedrlín, 1965)

For every $\kappa \geq \omega$, the **well-ordering number** of $\mathcal{G}^{\leq \kappa}$ with the **homomorphism** relation is $(2^\kappa)^+$.

Theorem (Komjáth, 1995)

For every $\kappa > \omega$, the **well-ordering number** of $\mathcal{G}^{\leq \kappa}$ with the **minor** relation is $(2^\kappa)^+$.

Corollary

The class of **all graphs** is **not κ -wqo** by [your favorite graph containment relation] for any κ .

Results of Shelah

Definition (Shelah, 1982)

Q is κ -wqo if in every sequence $\langle q_\alpha \mid \alpha < \kappa \rangle$ there are $\alpha < \beta < \kappa$ such that $q_\alpha \preceq q_\beta$.
The **well-ordering number** of Q is the least cardinal κ such that Q is κ -wqo (if it exists).

Theorem (Shelah, 1982)

The **well-ordering number** of order-theoretic **trees** with **injective level-embeddings** is a (mildly strong) **large cardinal**. In particular, the existence of such a cardinal cannot be proved in ZFC.

This suggests that ZFC **likely cannot prove** the existence of a cardinal number κ such that graph-theoretic **trees** would be κ -wqo by the **subgraph** relation.

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Definition (Shelah, 1982)

Q is κ -wqo if in every sequence $\langle q_\alpha \mid \alpha < \kappa \rangle$ there are $\alpha < \beta < \kappa$ such that $q_\alpha \preceq q_\beta$.
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This suggests that ZFC **likely cannot prove** the existence of a cardinal number κ such that graph-theoretic **trees** would be κ -wqo by the **subgraph** relation.

Definition

Q is **class-wqo** if in every sequence $\langle q_\alpha \mid \alpha \in \text{On} \rangle$ there are $\alpha < \beta$ such that $q_\alpha \preceq q_\beta$.

Problem

Is it possible to prove (without relying on any large cardinals) that graph-theoretic **trees** are **class-wqo** by the **subgraph** relation?

Vopěnka's principle

Axiom (Vopěnka's principle)

The category of graphs has no large discrete full subcategory.

Vopěnka's principle is weaker than a huge cardinal but it implies a proper class of measurable cardinals.

Theorem

Denote by $\mathcal{G}$ the class of all graphs. The following statements are equivalent to Vopěnka's principle:

- 1 $\mathcal{G}$ is *class-wqo* by the *homomorphism* relation.
- 2 $\mathcal{G}$ is *class-wqo* by the *subgraph* relation.
- 3 $\mathcal{G}$ is *class-wqo* by the *induced subgraph* relation.

Theorem (Komjáth, 1995)

$\mathcal{G}$ is *not* κ -wqo by the *minor* relation for any κ , no matter how large.

Problem

Vopěnka's principle clearly implies that $\mathcal{G}$ is class-wqo by the minor (or topological minor) relation. Does the converse implication hold as well?